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Please show all work clearly. Take care to communicate effectively and write mathematically sound work. If you have any questions, then please let me know.

1. Find the center,  $c$ , and radius,  $r$  of the sphere given by  $x^2 + y^2 + z^2 - 6x + 2z = 0$ .

4/5

$$x^2 + y^2 + z^2 - 6x + 2z = 0$$

$$x^2 - 6x + y^2 + z^2 + 2z = 0$$

$$(x^2 - 6x + 9) - 9 + y^2 + (z^2 + 2z + 1) - 1 = 0$$

$$(x - 3)^2 + y^2 + (z + 2)^2 = 10$$

Center:  $(3, 0, -2)$ .

Radius:  $\sqrt{10}$ 

2. Find the angle,  $\theta$ , between the planes  $M$  and  $N$ , where  $M$  is given by  $8x + 6y = -3$  and  $N$  is given by  $9x + 5y + 10z = -19$ . Give your answer in degrees, rounded to two decimal places.

6/6

$M: 8x + 6y = -3$   
 $\vec{n}_1 = \text{normal vector for } M$   
 $\vec{n}_1 = \langle 8, 6, 0 \rangle$

$N: 9x + 5y + 10z = -19$   
 $\vec{n}_2 = \text{normal vector for } N$   
 $\vec{n}_2 = \langle 9, 5, 10 \rangle$

$$\theta = \cos^{-1} \left( \frac{\vec{n}_1 \cdot \vec{n}_2}{\|\vec{n}_1\| \|\vec{n}_2\|} \right)$$

$$\vec{n}_1 \cdot \vec{n}_2 = (8)(9) + (6)(5) + (0)(10) = 102$$

$$= \cos^{-1} \left( \frac{102}{(10)(\sqrt{206})} \right)$$

$$= 44.71^\circ$$

3. Find the point,  $P(x, y, z)$ , at which the line,  $\ell$ , given by  $x(t) = 1 + 6t$ ,  $y(t) = 2 + 2t$ , and  $z(t) = 1 + 5t$ , where  $-\infty < t < \infty$ , intersects the plane,  $M$ , given by  $x + y + z = -9$ .

3/3

$$x(t) = 1 + 6t$$

$$y(t) = 2 + 2t$$

$$z(t) = 1 + 5t$$

$$x + y + z = -9$$

$$(1 + 6t) + (2 + 2t) + (1 + 5t) = -9$$

$$1 + 6t + 2 + 2t + 1 + 5t = -9$$

$$4 + 13t = -9$$

$$t = -1$$

$$x(-1) = 1 + 6(-1)$$

$$= -5$$

$$y(-1) = 2 + 2(-1)$$

$$= 0$$

$$z(-1) = 1 + 5(-1)$$

$$= -4$$

So, the line intersects the plane at  $(-5, 0, -4)$



## Exam 2

MAT 262

4. Consider the vectors  $\mathbf{v} = \langle 6, 3, -2 \rangle$  and  $\mathbf{u} = \langle 3, 4, 0 \rangle$ . Compute  $\text{proj}_{\mathbf{v}}(\mathbf{u})$ .

4/4

$$\vec{v} = \langle 6, 3, -2 \rangle$$

$$\vec{u} = \langle 3, 4, 0 \rangle$$

$$\vec{u} \cdot \vec{v} = (6 \cdot 3) + (3 \cdot 4) + (-2 \cdot 0) = 30$$

$$|\vec{v}| = 7$$

$$\text{proj}_{\vec{v}}(\vec{u}) = \left( \frac{\vec{u} \cdot \vec{v}}{|\vec{v}|^2} \right) \vec{v}$$

$$= \frac{30}{49} \langle 6, 3, -2 \rangle$$

$$= \left\langle \frac{180}{49}, \frac{90}{49}, -\frac{60}{49} \right\rangle$$

5. Find the distance from the point  $S(4, -2, 4)$  to the plane,  $M$ , given by  $x + 2y + 2z = 17$ .

6/6

$$S(4, -2, 4)$$

$$x + 2y + 2z = 17$$

$$P(17, 0, 0)$$

$$x + 2(0) + 2(0) = 17$$

$$x = 17$$

$$\vec{PS} = \langle -13, -2, 4 \rangle \quad (17, 0, 0)$$

$$\vec{n} = \langle 1, 2, 2 \rangle$$

$$|\vec{n}| = 3$$

$$\text{Distance} = \left| \vec{PS} \cdot \frac{\vec{n}}{|\vec{n}|} \right|$$

$$= \left| \frac{1}{3} (-9) \right|$$

$$= |-3|$$

$$= 3$$

where,  $S$  is the point in space and  $P$  is the point on the plane.

6. Determine the area,  $A$ , of the triangle with vertices at  $P(1, 3, -2)$ ,  $Q(-1, 2, 0)$ , and  $R(-1, 5, -1)$ .

4/5

$\vec{u}$  is the vector from  $P$  to  $Q$

$$= \vec{PQ}$$

$\vec{v}$  is the vector from  $P$  to  $R$

$$= \vec{PR}$$

$$\text{Area} = \frac{1}{2} |\vec{u} \times \vec{v}|$$

$$= \frac{1}{2} (\sqrt{65})$$

$$= \frac{\sqrt{65}}{2}$$

$$P(1, 3, -2)$$

$$Q(-1, 2, 0)$$

$$R(-1, 5, -1)$$

$$\vec{u} = \langle -2, -1, 2 \rangle$$

$$\vec{v} = \langle -2, 2, 1 \rangle$$

$$\vec{u} \times \vec{v} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -2 & -1 & 2 \\ -2 & 2 & 1 \end{vmatrix}$$

$$= \vec{i} \begin{vmatrix} -1 & 2 \\ 2 & 1 \end{vmatrix} - \vec{j} \begin{vmatrix} -2 & 2 \\ -2 & 1 \end{vmatrix} + \vec{k} \begin{vmatrix} -2 & -1 \\ -2 & 2 \end{vmatrix}$$

$$= \langle -5, -2, -6 \rangle$$

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7. Find the volume,  $V$ , of the parallelepiped determined by  $\vec{u} = \langle 4, 1, 0 \rangle$ ,  $\vec{v} = \langle 2, -1, 2 \rangle$ , and  $\vec{w} = \langle 4, 0, 4 \rangle$ .

3.5/4

$$\text{Volume} = |(\vec{u} \times \vec{v}) \cdot \vec{w}|$$

$$\vec{u} \times \vec{v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & 1 & 0 \\ 2 & -1 & 2 \end{vmatrix}$$

$$= \hat{i} \begin{vmatrix} 1 & 0 \\ -1 & 2 \end{vmatrix} - \hat{j} \begin{vmatrix} 4 & 0 \\ 2 & 2 \end{vmatrix} + \hat{k} \begin{vmatrix} 4 & 1 \\ 2 & -1 \end{vmatrix}$$

$$= \langle 2, 8, -6 \rangle$$

$$= |\langle 2, 8, -6 \rangle \cdot \langle 4, 0, 4 \rangle|$$

$$= 16$$

or

$$|(\vec{u} \times \vec{v}) \cdot \vec{w}| = \left| \begin{vmatrix} 4 & 1 & 0 \\ 2 & -1 & 2 \\ 4 & 0 & 4 \end{vmatrix} \right| \quad \text{abs. value!}$$

$$= \begin{vmatrix} 4 & 1 & 0 \\ 2 & -1 & 2 \\ 4 & 0 & 4 \end{vmatrix} = 4 \begin{vmatrix} 1 & 2 \\ 0 & 4 \end{vmatrix} - 1 \begin{vmatrix} 2 & 2 \\ 4 & 4 \end{vmatrix} + 0 \begin{vmatrix} 2 & -1 \\ 4 & 0 \end{vmatrix}$$

$$= 16$$

$$= 16$$

8. Find the distance,  $d$ , from the point  $S(1, 3, -2)$  to the line,  $\ell$ , given parametrically by  $x(t) = -1 - 2t$ ,  $y(t) = 5 - t$ , and  $z(t) = -1 + 2t$ , where  $-\infty < t < \infty$ .

6/6

$$\text{distance} = \frac{|\vec{PS} \times \vec{v}|}{|\vec{v}|}$$

$$S(1, 3, -2)$$

$$P(-1, 5, -1)$$

$$\vec{PS} = \langle 2, -2, -1 \rangle$$

$$\vec{v} = \langle -2, -1, 2 \rangle$$

$$|\vec{v}| = 3$$

where,  $S$  is a point in space and  $P$  is the point on the line.

$$= \frac{\sqrt{65}}{3}$$

$$|\vec{PS} \times \vec{v}| = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -2 & -1 \\ -2 & -1 & 2 \end{vmatrix}$$

$$= \hat{i} \begin{vmatrix} -2 & -1 \\ -1 & 2 \end{vmatrix} - \hat{j} \begin{vmatrix} 2 & -1 \\ -2 & 2 \end{vmatrix} + \hat{k} \begin{vmatrix} 2 & -2 \\ -2 & -1 \end{vmatrix}$$

$$= \langle -5, -2, -6 \rangle$$

$$= \sqrt{65}$$



9. Find a parametrization of the line,  $\ell$ , in which the planes  $M$  and  $N$  intersect, where  $M$  is given by  $x + 3y + 3z = -1$  and  $N$  is given by  $x + 2y + 2z = 0$ .

6.5/7  
 $M: x + 3y + 3z = -1$   
 Normal line for  $M$  is  $\vec{m}$   
 $\vec{m} = \langle 1, 3, 3 \rangle$

$N: x + 2y + 2z = 0$   
 Normal line for  $N$  is  $\vec{n}$   
 $\vec{n} = \langle 1, 2, 2 \rangle$

$\vec{v} = \vec{n} \times \vec{m}$

$$= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 2 & 2 \\ 1 & 3 & 3 \end{vmatrix}$$

$$= \vec{i} \begin{vmatrix} 2 & 2 \\ 3 & 3 \end{vmatrix} - \vec{j} \begin{vmatrix} 1 & 2 \\ 1 & 3 \end{vmatrix} + \vec{k} \begin{vmatrix} 1 & 2 \\ 1 & 3 \end{vmatrix}$$

$= \langle 0, -1, 1 \rangle$

Solving  $z = 0$

Equations

$$x + 3y = -1$$

$$x + 2y = 0$$

$$-3y - 1 + 2y = 0$$

$$-y - 1 = 0$$

$$-y = 1$$

$$y = -1$$

$$x + 3(-1) = -1$$

$$x - 3 = -1$$

$$x = 2$$

So,

$$x(t) = 2$$

$$y(t) = -1 - t$$

$$z(t) = t$$

Whole

$-\infty < t < \infty$

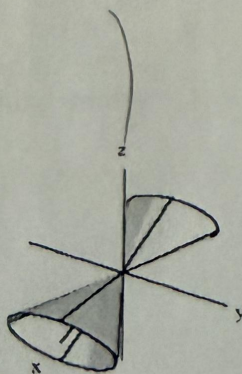
10. Draw lines to match each of the following equations to one of the given graphs. Then draw lines to match each graph to one of the given names.

0.5/4  
 $16y^2 + 49z^2 = x^2$

$$y^2 - x^2 = 4z$$

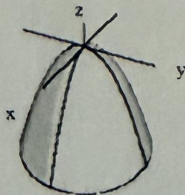
$$-\frac{x^2}{25} - \frac{y^2}{16} = z$$

$$z^2 - x^2 - y^2 = 64$$



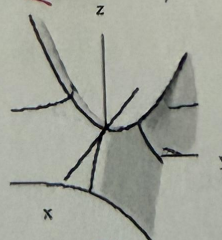
?

Elliptical  
Paraboloid



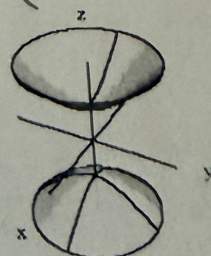
?

Elliptical  
Cone



?

Hyperboloid  
of Two Sheets



?

Hyperbolic  
Paraboloid